MTH 403 سیکھیں
VU 100% کامیابی آسان طریقے سے
Virtual University
Short Lecture
40,41 SPHERICAL TRIGONOMETRY
SIR NAIMAT ULLAH

Lecture 40

Spherical Trigonometry

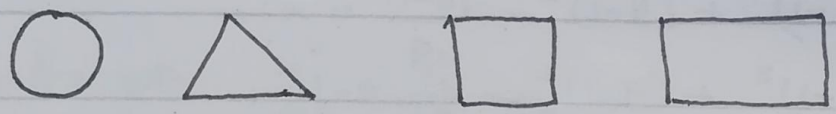
Main branch of mathematics

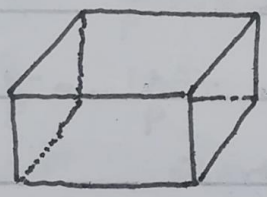
(1) Arithmetic (ii) Algebra (iii) Geometry

Geometry

the mathematical study of properties, measurement, relationship of points, circles, line planes, surfaces, angles etc.

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Two-dimensional figures

It is a figure that has length and width but no depth. In x-axis and y-axis

e.g. circles, squares, Rectangle, Triangles etc

Three-dimensional figures

It is a figures that has length, width and depth. x-axis, y-axis, z-axis

e.g. cube, ~~prime~~ prism, sphere, cylinder, cone.

Types of Geometry

(1) Plane geometry (ii) Solid geometry

(iii) Spherical geometry

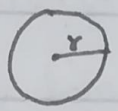
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1 Plane geometry

which deals with only two dimensions.

e.g circle

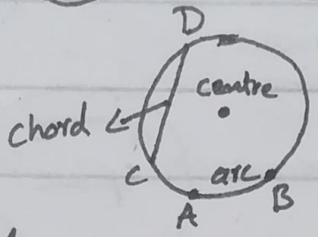


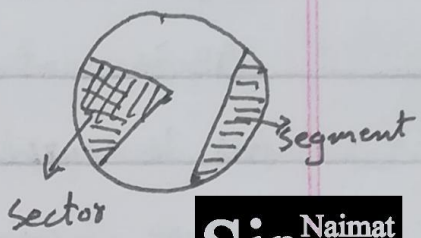


r = radius

d = diameter

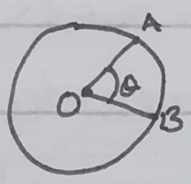
$d = 2r$





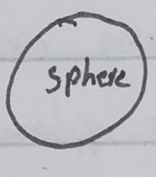
$C = 2\pi r = \pi d$

$A = \pi r^2$



l = length of arc $= r\theta$

Area of sector $= \frac{1}{2} r^2 \theta$



$A = 4\pi r^2$

$V = \frac{4}{3} \pi r^3$

(2) Solid Geometry →

which deals with all Three-dimensions.

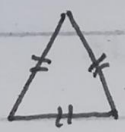
e.g. cube, sphere, cylinder, cone, etc.

Trignometry

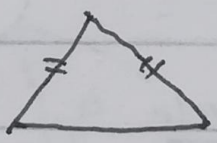
It is the branch of geometry that studies relationships involving length and angles of Triangles.

By side

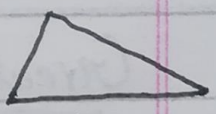
e.g



Equilateral △



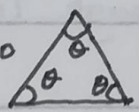
Isosceles △



scalene △

By angles

all angles < 90



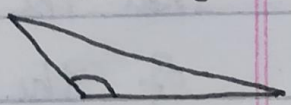
Acute △

one angle $= 90$



Right △

one angle > 90



Obtuse △

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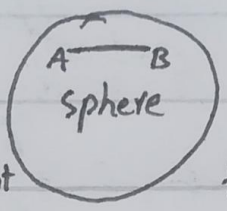
Spherical geometry

The study of figures on The surface of The sphere.

Surface of sphere is not flat

AB is not a straight line

AB is curve because surface of sphere is not flat



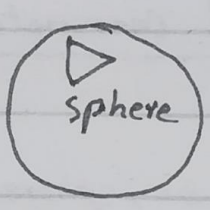
football

Spherical Trignometry

It deals with triangles drawn on The surface of sphere.

not plane triangle

all sides are arc

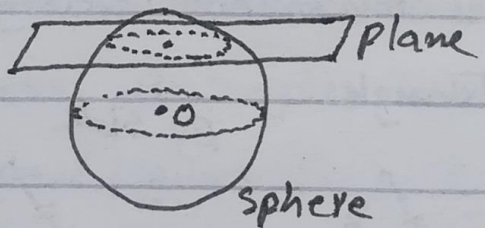


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Lecture – 41

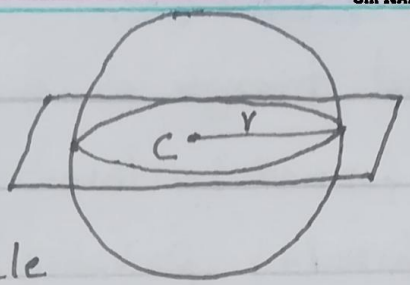
Intersection of a plane and a sphere

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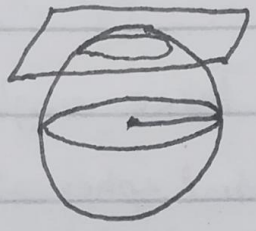
Great circle

A great circle of a sphere is The intersection of the sphere and a plane such that the plane passes Through the centre of the sphere.



Small Circle

A small circle of a sphere is the intersection of the sphere and a plane such that the plane does not pass through the centre of the sphere.



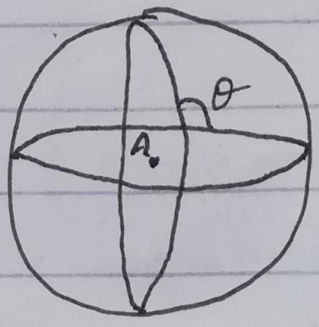
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Shortest distance between two points on the surface of sphere/earth

Spherical angle

It is the angle between two intersecting arcs on a sphere, and is measured by the angle between the planes of the great circles containing the arcs.

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Spherical Triangle

A spherical triangle is a part of surface of sphere formed by Three great circular arcs intersecting pairwise in Three vertices.

Spherical Triangle

1. The Three sides are all The arcs of great circles.
2. Any Two sides are Two together longer Than The Third side
3. The sum of Three angles is greater Than 180°
4. Each individual spherical angle is less Than 180°

Plane triangle

1. The Three sides are straight lines on The surface of plane
2. Any two sides are Together longer Than The Third side
3. The sum of Three angles is always equal to 180°
4. Each individual angle is less Than 180°

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